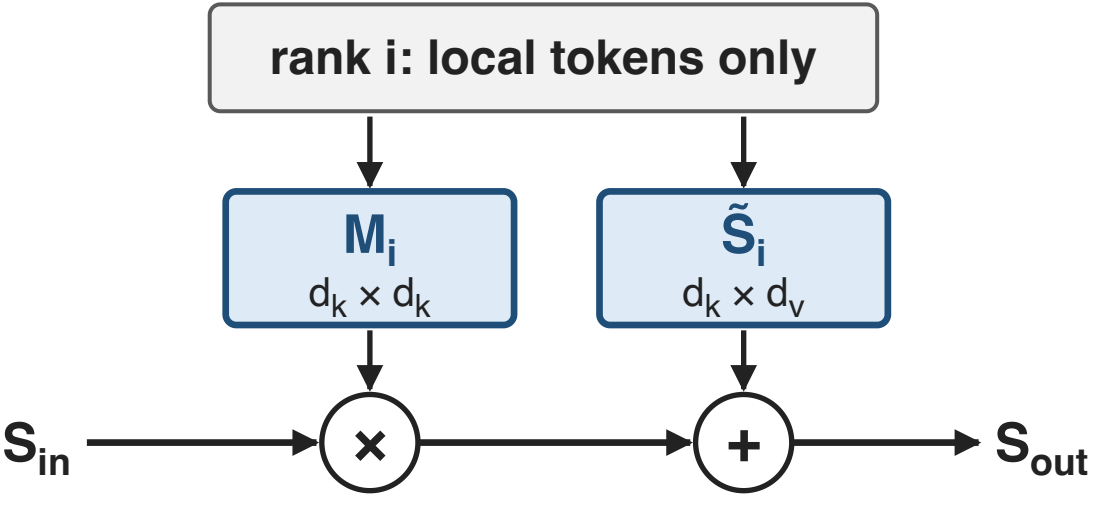

$$S_{out} = M_i \cdot S_{in} + \tilde{S}_i$$

affine maps compose: prefix scan